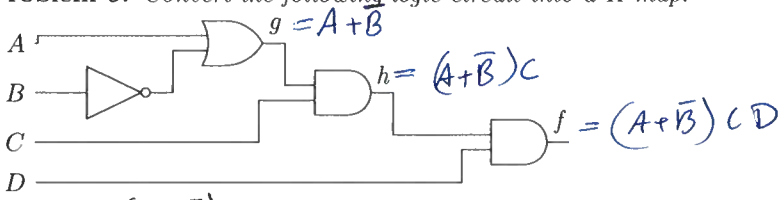
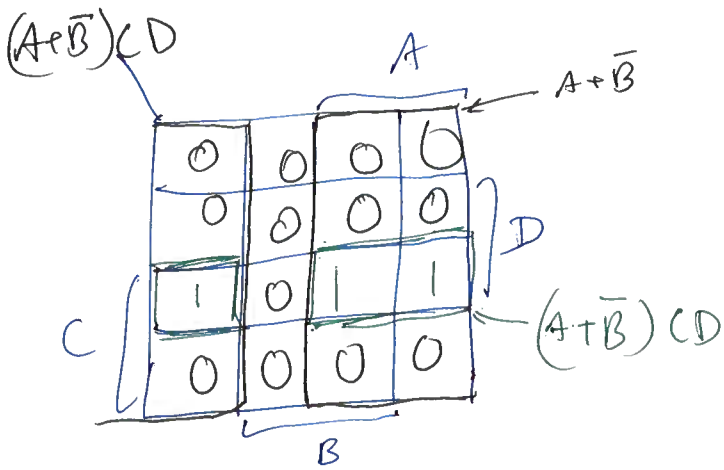
**Example 11.** *Convert the following Boolean expression into a K-map.* $f = \overline{A\bar{B} + CD}$

**Problem 9.** *Convert the following logic circuit into a K-map.*




## 1.9 Boolean Algebra

### 1.9.1 Axioms of Boolean algebra

1. $0 \cdot 0 = 0$
2. $1 + 1 = 1$

3. $1 \cdot 1 = 1$
4. $0 + 0 = 0$
5. $0 \cdot 1 = 1 \cdot 0 = 0$
6. $\bar{0} = 1$
7. $\bar{1} = 0$
8. $x = 0$ if $x \neq 1$
9. $x = 1$ if $x \neq 0$

### 1.9.2 Single variable theorems (Prove by drawing K-maps)

1. $x \cdot 0 = 0$

2. $x + 1 = 1$

3. $x \cdot 1 = x$

4. $x + 0 = x$

5. $x \cdot x = x$

6. $x + x = x$

7. $x \cdot \bar{x} = 0$

8. $x + \bar{x} = 1$

9. $\bar{\bar{x}} = x$

**Remark 2** (Duality). *Swap + with · and 0 with 1 to get another theorem*

### 1.9.3 Two and three variable properties (Prove by K-maps)

1. Commutative: $x \cdot y = y \cdot x$ , $x + y = y + x$

2. Associative: $x \cdot (y \cdot z) = (x \cdot y) \cdot z$, $x + (y + z) = (x + y) + z$

3. Distributive: $x \cdot (y + z) = x \cdot y + x \cdot z$, $x + y \cdot z = (x + y) \cdot (y + z)$

4. Absorption: $x + x \cdot y = x$, $x \cdot (x + y) = x$

5. Combining: $x \cdot y + x \cdot \bar{y}$, $(x + y) \cdot (x + \bar{y}) = x$

6. DeMorgan's theorem: $\overline{x \cdot y} = \bar{x} + \bar{y}$, $\overline{x + y} = \bar{x} \cdot \bar{y}$.

7. Concensus:

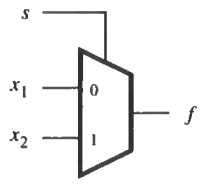
   (a) $x + \bar{x} \cdot y = x + y$

   (b) $x \cdot (\bar{x} + y) = x \cdot y$

   (c) $x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$

   (d) $(x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$

**Example 12** (Multiplexer). *Multiplexer is a circuit used to select one of the input lines $x_1$ and $x_2$ based only select input $s$. When $s = 0$, $x_1$ is selected, $x_2$ is selected otherwise. Find a boolean expression and a circuit for multiplexer*




| $s$ | $f(s, x_1, x_2)$ |
|---|---|
| 0 | $x_1$ |
| 1 | $x_2$ |

**Example 13.** *Simplify $f = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}$ using boolean algebra.*

**Problem 10** (30 marks, Exercise 2.14 [1]). *Simplify the following Boolean equations using Boolean theorems.*

$$Y = \bar{A}BC + \bar{A}B\bar{C} = \bar{A}B(C+\bar{C}) = \bar{A}B \tag{1.1}$$

$$Y = \overline{ABC} + A\bar{B} = (\bar{A}+\bar{B}+\bar{C}) + A\bar{B} = \bar{A} + (1+A)\bar{B} + \bar{C} = \bar{A} + \bar{B} + \bar{C} \tag{1.2}$$

$$Y = ABC\bar{D} + A\overline{BCD} + (\overline{A+B+C+D}) \tag{1.3}$$

$$= ABC\bar{D} + A(\bar{B}+\bar{C}+\bar{D}) + \bar{A}\bar{B}\bar{C}\bar{D}$$

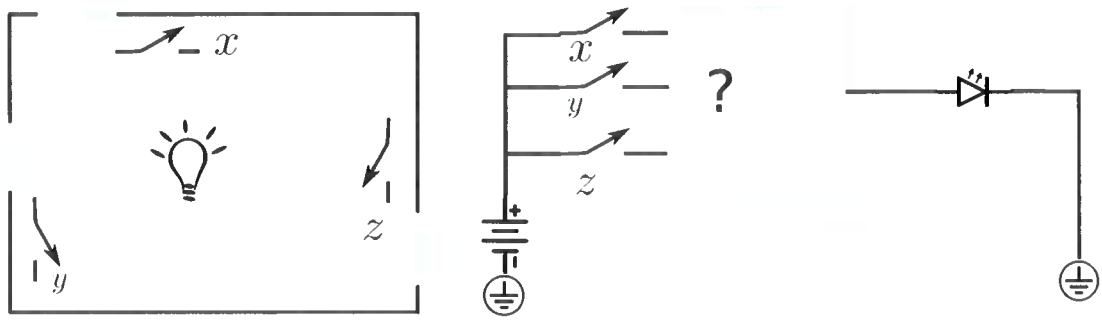
**Example 14.** *Simplify $f = \bar{A}\bar{A}\bar{C} + \bar{A}\bar{B}C$ using K-maps.*

$$= \underbrace{ABC\bar{D} + A\bar{B} + A\bar{C} + A\bar{D}} + A\bar{B}\bar{C}\bar{D}$$

$$= A(BC+1)\bar{D} + A\bar{B} + A\bar{D} + A\bar{B}\bar{C}\bar{D}$$

$$= A\bar{D} + A\bar{B} + A\bar{D} + A\bar{B}\bar{C}\bar{D}$$

**Example 15.** *Assume that a large room has three doors and that a switch near each door controls a light in the room. It has to be possible to turn the light on or off by changing the state of any one of the switches.*




**Problem 11** (20 marks, Exercise 2.38 [1]). *An M-bit thermometer code for the number k consists of k 1's in the least significant bit positions and M − k 0's in all the more significant bit positions. A binary-to-thermometer code converter has N inputs and $2^{N-1}$ outputs. It produces a $2^N-1$ bit thermometer code for the number specified by the input. For example, if the input is 110, the output should be 0111111. Design a 3:7 binary-to-thermometer code converter. Give a simplified Boolean equation for each output.*

| Inputs | | | Outputs | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $I_2$ | $I_1$ | $I_0$ | $O_6$ | $O_5$ | $O_4$ | $O_3$ | $O_2$ | $O_1$ | $O_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$O_0 = M_0 = I_2 + I_1 + I_0$

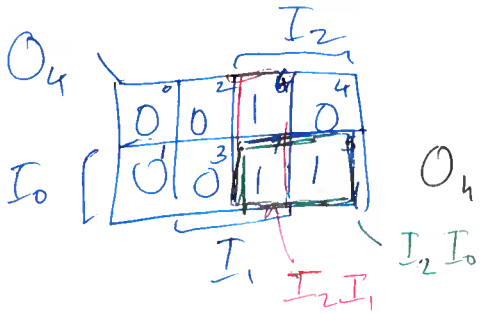
$O_6 = m_7 = I_2 I_1 I_0$

$O_1 = I_2 + I_1$ because combined region of $I_2$ and $I_1$

$O_3 = I_2$ because same activations as $I_2$

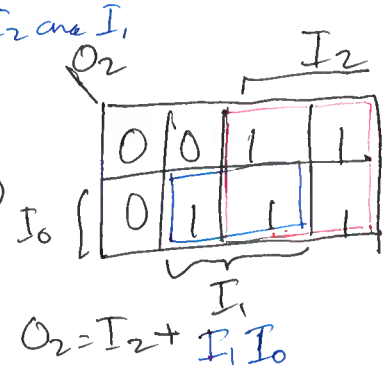
$O_5 = I_2 \cdot I_1$ because common region of $I_2$ and $I_1$

$O_4$




$O_4 = I_2(I_1 + I_0)$

$O_2$




$O_2 = I_2 + I_1 I_0$

# Bibliography

[1] Sarah L Harris and David Harris. *Digital design and computer architecture.* Morgan Kaufmann, 2022.

[2] Brown Stephen and Vranesic Zvonko. *Fundamentals of digital Logic with Verilog design.* McGraw Hill, 2022.

# Chapter 2

# Logic minimization

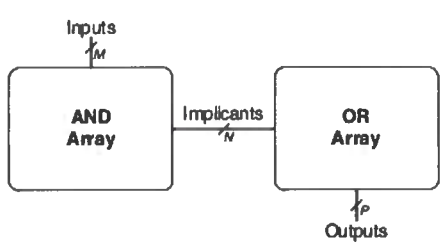
## 2.1 Logic minimization

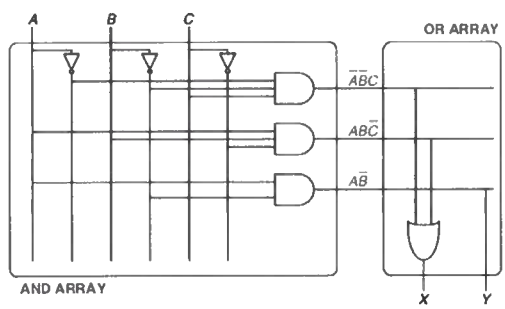
A general optimization criteria for multi-level logic are to Minimize some combination of:

1. Area occupied by the logic gates and interconnect;
2. the Critical Path Delay of the longest path through the logic;
3. the Degree of Testability of the circuit, measured in terms of the percentage of faults covered by a specified set of test vectors, for an appropriate fault model (Eg., single stuck faults, multiple stuck faults, etc.);
4. Power consumed by the logic gates.

In this course, we will start with two-level multi-input circuits and a criteria based on the number of gates/transistors/diodes.

## 2.2 Programmable Logic Arrays




## 2.3 Two-level circuits

The cost that we are going to consider in this class depend upon:

1. Number of gates.
2. Number of input to the gates.

More gates need more transistors, more area on the chip. More-inputs the gate need more transistors within each gate. Number of gate inputs can be considered secondary criterion to the number of gates.

**Example 2.1.** *Find the cost of the following Boolean expression* $X = \bar{A}\bar{B}C + AB\bar{C} + A\bar{B}$. 1 gate + 3 inputs

**Problem 2.1** (5 marks). *Find the cost of the following Boolean expression* $X = A\bar{B}C + \bar{A}B\bar{C} + \bar{B}C$.

1 gate + 3 inputs 1 gate + 3 inputs 1 gate + 2 inputs

= 3 AND gates + 8 inputs + 1 OR gate + 3 inputs = 15

## 2.4 Terminology for K-maps

Running Example: $f = \sum m(0, 1, 2, 3, 7) = \bar{x}_1 + x_1x_2x_3$.

**Literal** A single variable or its complement. Example: $\bar{x}, x_1, x_2, x_3$

**Implicant** A product term which is true for a function. All minterms are implicants. Example: $x_1x_2x_3$, $\bar{x}_1$, $m_0 = \bar{x}_1\bar{x}_2\bar{x}_3$, $\bar{x}_1x_3$, $\bar{x}_1\bar{x}_3$.

**Prime Implicant** An implicant that cannot be combined into fewer literals. Example: $\bar{x}_1, x_2x_3$.

**Essential Prime Implicant** An implicant that cannot be combined into fewer literals. Example: $x_2x_3$.

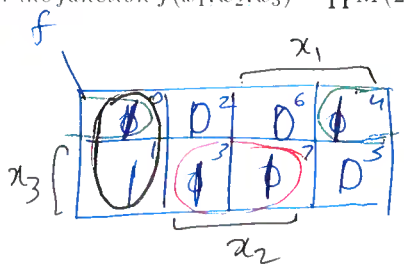
**Cover** : List of Prime Implicants that account for all $f = 1$.

**Cost** : Number of gates (excluding not gate on literals) and number of inputs to each gate.

**Example 2.2.** *Find minimum cost expression for the function* $f(x_1, x_2, x_3) = \prod M(4, 5, 6)$

**Problem 2.2** (10 marks). *Find minimum cost expression for the function* $f(x_1, x_2, x_3) = \prod M(2, 5, 6)$

$f = \bar{x}_1\bar{x}_2 + x_2x_3 + \bar{x}_2\bar{x}_3$




OR

$\bar{f} = x_2\bar{x}_3 + x_1\bar{x}_2x_3$

$f = (\bar{x}_2 + x_3) \cdot (\bar{x}_1 + x_2 + x_3)$

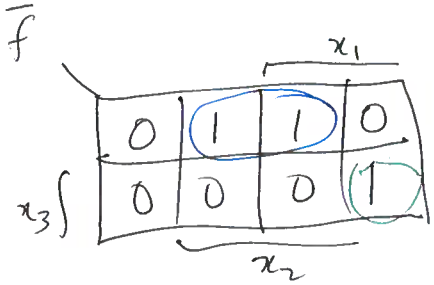


Figure 2.1: 7 Segment Representations of Each Integer

### 2.4.1 Incompletely specified functions or Don't cares

| BCD Value | | | | LED Segment |
|---|---|---|---|---|
| $D_3$ | $D_2$ | $D_1$ | $D_0$ | E |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | d |
| 1 | 0 | 1 | 1 | d |
| 1 | 1 | 0 | 0 | d |
| 1 | 1 | 0 | 1 | d |
| 1 | 1 | 1 | 0 | d |
| 1 | 1 | 1 | 1 | d |

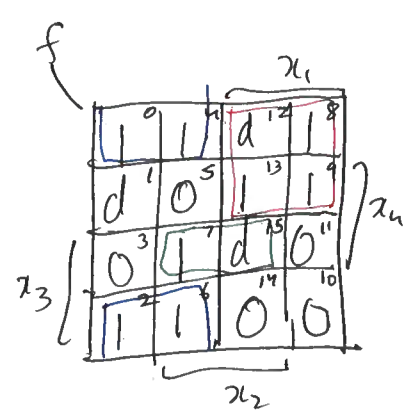
**Example 2.3.** *Find minimum cost expression for the function*

$$f(x_1, \ldots, x_4) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$$

**Problem 2.3** (10 marks). *Find minimum cost expression for the function*

$$f(x_1, \ldots, x_4) = \sum m(0, 2, 4, 6, 7, 8, 9, 13) + D(1, 12, 15)$$

$$f = x_1 \bar{x}_3 + \bar{x}_1 \bar{x}_4 + x_2 x_3 x_4$$



## 2.5 A few more Boolean problems

**Example 2.4.** *Simplify the following Boolean expression:*

$$f = x_1\bar{x}_3\bar{x}_4 + x_2\bar{x}_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_3$$

**Problem 2.4** (20 marks). *A simple security system for two doors consists of a card reader and a keypad.*

*A person may open a particular door if he or she has a card containing the corresponding code and enters an authorized keypad code for that card. Note that card-code and keypad-code are different. The outputs from the card reader are given in the table below.*

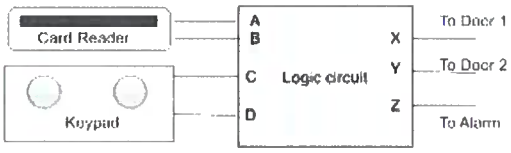
*To unlock a door, a person must hold down the proper keys on the keypad and, then, insert the card in the reader. The authorized keypad code for door 1 is 10, and the authorized keypad code for door 2 is 11. If the card has an invalid code or if the wrong keypad code is entered, the alarm will ring when the card is inserted. If the correct keypad code is entered, the corresponding door will be unlocked when the card is inserted.*

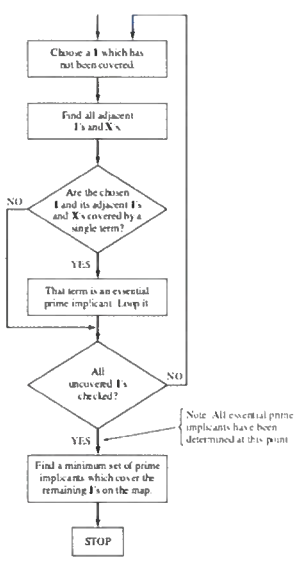
*Design the logic circuit for this simple security system. Your circuit's inputs will consist of a card code AB, and a keypad code CD. The circuit will have three outputs XYZ (if X is 1, door 1 will be opened; if Y is 1, door 2 will be opened; if Z 1, the alarm will sound).*

*Find the minimal cost two-level circuit using K-maps for X, Y, Z. Provide the minimal cost. (It can be either of SOP/POS forms)*

| Card code | A | B |
|---|---|---|
| No card inserted | 0 | 0 |
| Valid card-code for door 1 | 0 | 1 |
| Valid card-code for door 2 | 1 | 1 |
| Invalid card code | 1 | 0 |

← Nothing happens




MSB → LSB

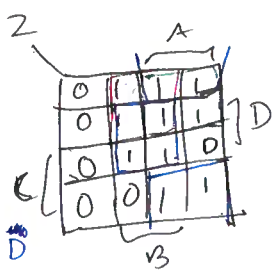
| A | B | C | D | X | Y | Z |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 |

D1 D2 ← Alarm

Nothing happens

← correct card code and keypad code for door 1

← for door 2



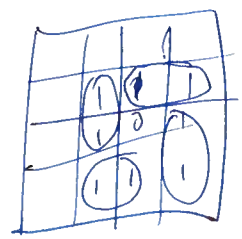

$X = m_6 = \bar{A}BC\bar{D}$, $Y = m_{11} = A\bar{B}CD$, $Z = B\bar{C} + A\bar{C} + A\bar{D} + B\bar{D}$

| Row | $x_1$ | $x_2$ | $x_3$ | f |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 2 | 0 | 1 | 0 | 1 |
| 3 | 0 | 1 | 1 | 0 |
| 4 | 1 | 0 | 0 | 1 |
| 5 | 1 | 0 | 1 | 0 |
| 6 | 1 | 1 | 0 | 0 |
| 7 | 1 | 1 | 1 | 1 |

Table 2.1: Truth table for a 3-way light switch

**Example 2.5.**

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | d | 1 | | 1 |
| 01 | | 1 | 1 | 1 |
| 11 | | d | d | |
| 10 | | 1 | | 1 |



**Problem 2.5** (10 marks). *Find the minimum SOP (sum of products) and POS (product of sum) expression for the function* $f(a,b,c,d) = \prod M(5,7,13,14,15) \cdot \prod D(1,2,3,9)$

**Problem 2.6** (10 marks). *If the Sum of Products (SOP) form for* $\bar{f} = AB\bar{C} + \bar{A}\bar{B}$, *then give the Product of Sums (POS) form for* $f$.

**Problem 2.7** (10 marks). *Use DeMorgan's Theorem to find* $f$ *if* $\bar{f} = (A + \bar{B}C)D + EF$.

**Example 2.6.** *Implement the function in Table 2.1 using only NAND gates.*

**Problem 2.8** (10 marks). *Implement the function in Table 2.1 using only NOR gates.*

**Problem 2.9** (10 marks). *Find the minimum-cost Sum of Products (SOP) and Product of Sums (POS) forms for the function* $f(x_1,x_2,x_3) = m(1,3,4,5)$. *Chose the minimum-cost expression by comparing Product of Sums (POS) and Sum of Products (SOP) forms.*

**Problem 2.10** (10 marks). *Find the minimum-cost Sum of Products (SOP) and Product of Sums (POS) forms for the function* $f(x_1,x_2,x_3) = \sum m(1,5,7) + D(2,4)$.

**Problem 2.11** (10 marks). *Find the minimum-cost Sum of Products (SOP) and Product of Sums (POS) forms for the function* $f(x_1,x_2,x_3,x_4) = \prod M(1,2,4,5,7,8,9,10,12,14,15)$. *Chose the minimum-cost expression by comparing Product of Sums (POS) and Sum of Products (SOP) forms.*

**Problem 2.12** (10 marks). *Derive a minimum-cost realization of the four-variable function that is equal to 1 if exactly two or exactly three of its variables are equal to 1; otherwise it is equal to 0.*

# Bibliography

[1] Sarah L Harris and David Harris. *Digital design and computer architecture.* Morgan Kaufmann, 2022.

[2] Brown Stephen and Vranesic Zvonko. *Fundamentals of digital Logic with Verilog design.* McGraw Hill, 2022.

(2.5) $f(a,b,c,d) = \prod M(5,7,13,14,15) \cdot \prod D(1,2,3,9)$

$f$

| | | | |
|---|---|---|---|
| 1 (0) | 1 (4) | 1 (12) | 1 (8) |
| d (1) | 0 (5) | 0 (13) | d (9) |
| d (3) | 0 (7) | 0 (15) | 1 (11) |
| d (2) | 1 (6) | 0 (14) | 1 (10) |

a, b, c, d

SOP

~~f=~~

SOP $f = \bar{c}\bar{d} + a\bar{b} + \bar{a}\bar{d}$

$\bar{f}$

| | | | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| d | 1 | 1 | d |
| d | 1 | 1 | 0 |
| d | 0 | 1 | 0 |

a, b, c, d

$\bar{f} = bd + abc$

$f = (\bar{b}+\bar{d}) \cdot (\bar{a}+\bar{b}+\bar{c})$

(2.6) $\bar{f} = AB\bar{C} + \bar{A}\bar{B}$

$f = (\bar{A}+\bar{B}+C) \cdot (A+B)$ By DeMorgan's Theorm

(2.7) $\bar{f} = (A+\bar{B}C)D + EF$

$f = \overline{[(A+\bar{B}C)D]} \cdot [\overline{EF}]$

$= (\overline{A+\bar{B}C} + \bar{D}) \cdot (\bar{E}+\bar{F})$

$= (A \cdot (B+\bar{C}) + \bar{D}) \cdot (\bar{E}+\bar{F})$

(2.8) To get NOR-NOR circuit we have to find POS ~~expre~~ expression

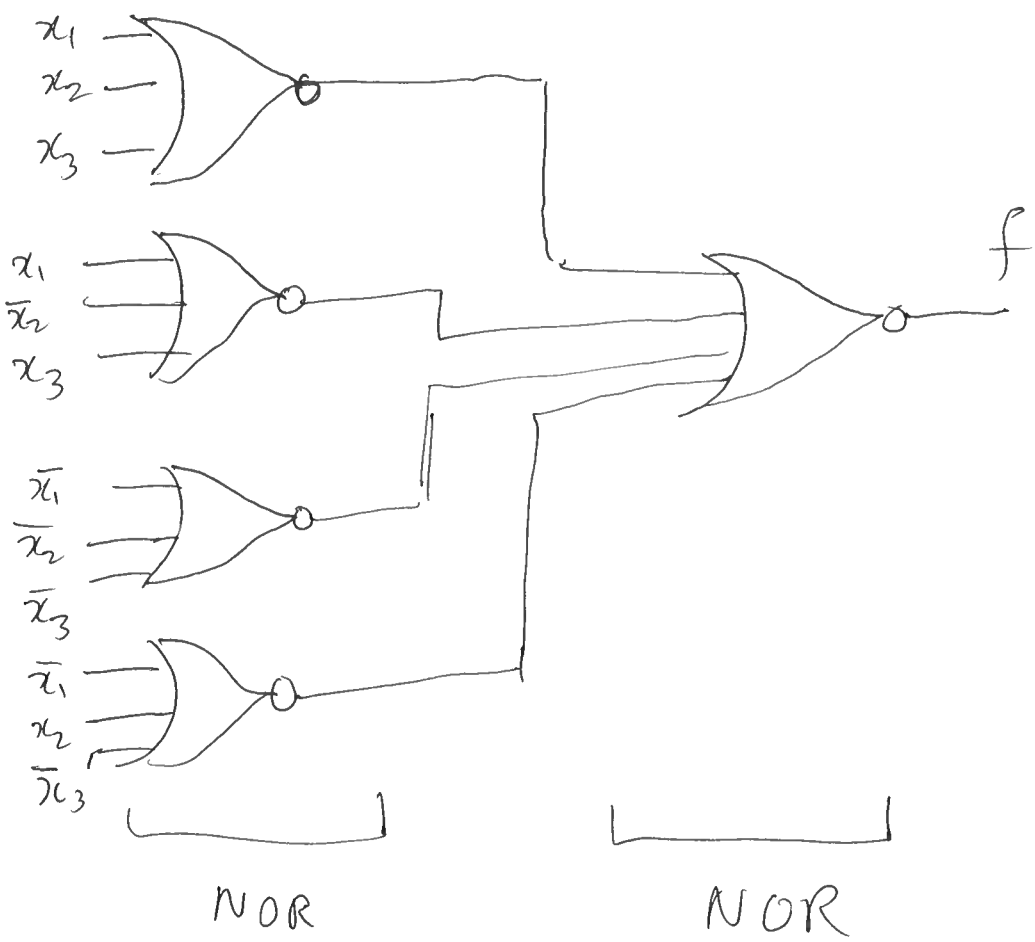
$\bar{f}$

| | | | | |
|---|---|---|---|---|
| | | | $x_1$ | |
| | 1 (0) | 0 (2) | 1 (6) | 0 (4) |
| $x_3$ | 0 (1) | 1 (3) | 0 (7) | 1 (5) |
| | | $x_2$ | | |

$$\bar{f} = m_0 + m_3 + m_6 + m_5$$

$$f = M_0 \cdot M_3 \cdot M_6 \cdot M_5 \leftarrow \text{POS expression}$$

$$f = (x_1 + x_2 + x_3) \cdot (x_1 + \bar{x}_2 + x_3) \cdot (\bar{x}_1 + \bar{x}_2 + x_3) \cdot (\bar{x}_1 + x_2 + \bar{x}_3)$$



## 2.9

$$f(x_1, x_2, x_3) = \sum m(1, 3, 4, 5)$$

$f$

| $x_3 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 (0) | 0 (2) | 0 (6) | 1 (4) |
| 1 | 1 (1) | 1 (3) | 0 (7) | 1 (5) |

$\bar{f}$

| $x_3 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 (0) | 1 (2) | 1 (6) | 0 (4) |
| 1 | 0 (1) | 0 (3) | 1 (7) | 0 (5) |

SOP $f = x_1 \bar{x}_2 + \bar{x}_1 x_3$

cost = 6

POS

$\bar{f} = \bar{x}_1 \bar{x}_3 + x_1 x_2$

$f = (x_1 + x_3) \cdot (\bar{x}_1 + \bar{x}_2)$

cost = 6

can choose either POS/SOP same cost.

## 2.10

$$f(x_1, x_2, x_3) = \sum m(1, 5, 7) + \sum D(2, 4)$$

$f$

| $x_3 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 (0) | d (2) | 0 (6) | d (4) |
| 1 | 1 (1) | 0 (3) | 1 (7) | 1 (5) |

$\bar{f}$

| $x_3 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 (0) | d | 1 | d |
| 1 | 0 (1) | 1 | 0 | 0 |

SOP

$f = \bar{x}_2 x_3 + x_1 x_3$

POS

$\bar{f} = \bar{x}_1 x_2 + \bar{x}_3$

$f = (x_1 + \bar{x}_2) \cdot x_3$ ✓

## 2.11

$$f(x_1, x_2, x_3, x_4) = \prod M(1, 2, 4, 5, 7, 8, 9, 10, 12, 14, 15)$$

$f$

| $x_3 x_4 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 (0) | 0 (4) | 0 (12) | 0 (8) |
| 01 | 0 (1) | 0 (5) | 1 (13) | 0 (9) |
| 11 | 1 (3) | 0 (7) | 0 (15) | 1 (11) |
| 10 | 0 (2) | 1 (6) | 0 (14) | 0 (10) |

SOP: cost = 23

$f = \bar{x}_2 x_3 x_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 + \bar{x}_1 x_2 x_3 \bar{x}_4 + x_1 x_2 \bar{x}_3 x_4$

SOP WINS

$\bar{f}$

| $x_3 x_4 \backslash x_1 x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 1 |
| 01 | 1 | 1 | 0 | 1 |
| 11 | 0 | 1 | 1 | 0 |
| 10 | 1 | 0 | 1 | 1 |

cost = 25

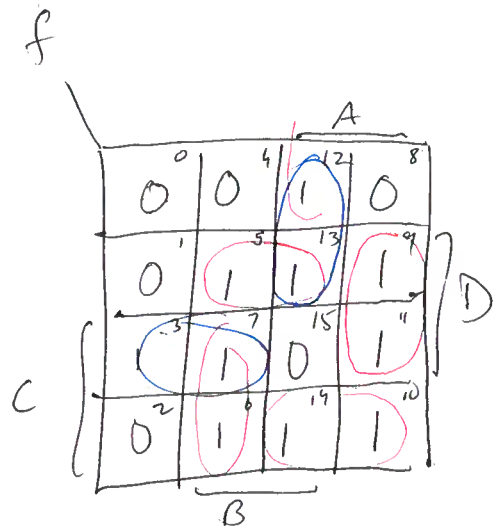
POS:

$\bar{f} = \bar{x}_2 x_3 \bar{x}_4 + \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_4 + x_2 x_3 x_4 + \bar{x}_2 \bar{x}_3 x_4$

$f = (x_2 + \bar{x}_3 + x_4) \cdot (x_1 + \bar{x}_2 + x_3) \cdot (\bar{x}_1 + x_4) \cdot (\bar{x}_2 + \bar{x}_3 + \bar{x}_4)(x_2 + x_3 + \bar{x}_4)$

2.12

| A | B | C | D | f |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 |




$f = \bar{A}CD + AB\bar{C} + A\bar{B}D + AC\bar{D}$
$+ \bar{A}BC + B\bar{C}D$